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VIEWING APPARATUS AND METHOD FOR PROJECTING A LIGHT SIGNAL

Abstract

A viewing apparatus is provided that has a first optical line having: a projecting system adapted to project a light signal on a first area of the retina of an eye of a wearer of the viewing apparatus, a light signal generator generating the light signal to be projected, and a waveguide having an input coupler and an exit coupler, the waveguide being adapted to convey light from the input coupler to the exit coupler, and optionally a second optical line having an optical system adapted to project a light signal of the environment on a second area of the retina of the eye of the wearer, the optical system comprising a part of the waveguide.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of U.S. application Ser. No. 17/607,113, having a filing date of Oct. 28, 2021, which is a national phase entry under 35 USC § 371 of International Application PCT/EP2020/061898, filed Apr. 29, 2020, and which claims priority to European patent application EP 19305561.3 filed May 2, 2019, all of said applications being incorporated herein by reference.

TECHNICAL FIELD OF THE INVENTION

[0002] The present invention concerns a viewing apparatus and a method for projecting light signal.

BACKGROUND OF THE INVENTION

[0003] The solution aims at stimulating retinal ganglion cells in the context of an optogenetics therapy.

[0004] The retina is composed of photoreceptors, which are highly specialized neurons that are responsible for photosensitivity of the retina by phototransduction, i.e. the conversion of light into electrical and chemical signals that propagate a cascade of events within the visual system, ultimately generating a representation of world. In the vertebrate retina, phototransduction is initiated by activation of light-sensitive receptor protein, rhodopsin.

[0005] Photoreceptor loss or degeneration, such as in case of retinitis pigmentosa (RP) or macular degeneration (MD), severely compromises, if not completely inhibits, phototransduction of visual information within the retina. Loss of photoreceptor cells and/or loss of a photoreceptor cell function are the primary causes of diminished visual acuity, diminished light sensitivity, and blindness.

[0006] Several therapeutic approaches dedicated to retinal degenerative diseases are currently in development, including gene therapy, stem cell therapy, optogenetics, and retinal prostheses.

[0007] For example it has been proposed to restore photosensitivity of the retina of a subject by controlling activity of defined populations of neurons without affecting other neurons in the brain by gene- and neuroengineering technology termed optogenetics. In contrast to traditional gene therapy that attempts to replace or repair a defective gene or bypass the genetic defect through correction of the protein deficiency or dysfunction, optogenetic approaches can be used to endow normally non-photosensitive cells in the retina with the ability to respond to light, thus restoring useful vision to the patient. Unlike retinal chip implants that provide extracellular electrical stimulation to bipolar or ganglion cells, optogenetics-based therapies stimulate the cells from inside the cell.

[0008] Optogenetics refers to the combination of genetics and optics to control well-defined events within specific cells of living tissue. Optogenetics consists in (i) genetically modifying target cells in order to render them sensitive to light by the expression of exogenous photoreactive proteins in cellular membrane and (ii) providing illuminating device able to provide light to said photoreactive proteins.

[0009] Examples of exogenous photoreactive proteins are provided in WO2007024391, WO2008022772 or WO2009127705 which describe the use of opsin genes derived from plants and microbial organisms (e.g. archaebacteria, bacteria, and fungi) encoding light-activated ion channels

and pumps (e.g. channelrhodopsin-2 [ChR2]; halorhodopsin [NpHR]), engineered for expression in mammalian neurons and which can be genetically targeted into specific neural populations using viral vectors. When exposed to light with appropriate wavelength, action potentials can be triggered in opsin-expressing neurons conferring thereby light sensitivity to these cells. Similarly, WO2013071231 discloses new channelrhodopsins, Chronos and Chrimson, which have different activation spectra from one another and from the state of the art (e.g., ChR2/VChR1), and allow multiple and distinct wavelengths of light to be used to depolarize different sets of cells in the same tissue, by expressing channels with different activation spectra genetically expressed in different cells, and then illuminating the tissue with different colors of light. The photoreactive protein disclosed in WO2017187272 is another alternative.

[0010] Optogenetics is an extremely powerful tool for selective neuronal activation/inhibition which can, for example, be used to restore neural functions in living animals, including humans, particularly in the eye.

[0011] Nevertheless, it has been shown that selected wavelengths of light shall be close to the optimal wavelengths of the photoreactive proteins and that these photoreactive proteins have a very low sensitivity to light. Therefore in order to obtain minimum level of protein activation by light, the intensity of light received by the target cell or protein shall be above a minimum value. As a consequence, an external device providing sufficient irradiance at the right wavelength is mandatory.

[0012] Alternatively, it has been proposed to restore at least partially vision in these patients with visual prosthesis systems. These systems are comprising a retina implant and are helpful tools for at least partially re-establishing a modest visual perception and a sense of orientation for blind and visually impaired users by exploiting said fact that although parts of the retinal tissue have degenerated most of the retina may remain intact and may still be stimulated directly by light dependent electrical stimuli. Typically, retina implant is implanted into the patient's eye, effecting electrical excitation of the remaining neuronal cells upon light stimulation. When being stimulated, these remaining neuronal cells convey the artificially induced electrical impulses to the visual part of the brain through the optic nerve.

[0013] Retinal implants can be broadly divided into two categories: epi- and sub-retinal. Epi-retinal devices are placed on or near the inner surface of the retina, i.e. the side of the retina which is first exposed to incident light and along which the nerve fibers of the ganglion cells pass on their way to the optic nerve. Epi-retinal implants typically comprise a chip with a plurality of pixel elements capable of receiving an image projected by an extraocular device (typically a camera and a microelectronic circuit for decoding incident light) on the retina through the lens of the eye, for converting the image into electrical signals and for further conveying the signals into electrical stimuli via a plurality of stimulation electrodes to stimulate the retinal cells adjacent the chip, in order to reconstruct or improve vision of blind or partially blind patients. In contrast, sub-retinal devices are placed under the retina, between the retina and the underlying retinal pigment epithelium or other deeper tissues. Currently available sub-retinal technologies rely on the implantation of a single, rigid and typically planar chip. It has been further shown that it is desirable to be able to implant more than one chip in order to cover a large visual field.

[0014] Retinal prostheses and optogenetic therapies rely on two main components. The first component engineered on the retina provides light sensitivity by providing a change of membrane potential of target retina cells: it is the implant in retinal prosthesis system or the light-gated ion channel protein genetically introduced in the retinal cells in optogenetic therapies. A second component is required to encode visual information (usually acquired with a camera or array of photodiodes) and to translate it in an input signal required by the former component. In retinal prostheses, the input signal is an electrical current delivered by a matrix of active electrodes or a pulse of light capable of activating passive components. In optogenetic gene therapy, the input signal which is delivered is a pulse of light at the appropriate intensity and wavelength required to

activate the optogenetic protein in a defined spatio-temporal manner.

[0015] The irradiance needed of the light cannot be achieved with standard headsets.

[0016] In particular, known solutions comprises projector to project images on the retina but it doesn't allow a field of view of more than 10° which is insufficient in some cases.

[0017] Moreover, heat produced by the required powerful light source and the control electronics induces a temperature increase that has to be kept within the bounds set by the standard ISO 60601-1 for Medical Electrical Equipment. The device shall also guarantee electrical safety (ISO 60601-1) and electro-magnetic compatibility (ISO 60601-1-2, ISO 60601-1-9).

[0018] Furthermore, an increase of temperature and of heat transfer of the device in front of the eye may induce dry eye and discomfort. Thus, further temperature constraints are required on the part of the device next to the eye.

[0019] It is desirable to provide a viewing apparatus able to deliver a larger field of view with a sufficient light intensity, while maintaining a sufficient level of comfort, particularly in terms of temperature for the wearer.

[0020] More specifically, it is desirable to provide a viewing apparatus able to limit excessive thermal energy from the light source and the control electronics and/or to facilitate dissipation of heat, so as to prevent a temperature of the illuminating device from going beyond the maximum temperature of normal operation thereof and providing an enlarged field of view.

[0021] It is further desirable to provide an illuminating device which is miniaturized so that it can be worn by humans on a daily basis.

SUMMARY OF THE INVENTION

[0022] There is therefore a need for a viewing apparatus adapted to project light signal on an eye of the wearer of the apparatus which enables to obtain an enlarged field of view, being more compact and providing an improved comfort and perception for the wearer.

[0023] To the end, the specification proposes a viewing apparatus comprising a first optical line comprising (i) a projecting system adapted to project a first light signal on a first area of the retina of an eye of a wearer of the viewing apparatus, (ii) a light signal generator generating the said first light signal to be projected and (iii) a waveguide comprising an input coupler and an exit coupler, the waveguide being adapted to convey light from the input coupler to the exit coupler.

[0024] According to preferred embodiment, the viewing apparatus further comprises a second optical line comprising an optical system adapted to project a second light signal of the environment (i.e. environment light signal) towards a second area of the retina of the eye of the wearer.

[0025] According to further aspects of the invention which are advantageous but not compulsory, the viewing apparatus might incorporate one or several of the following features, taken in any technically admissible combination: [0026] the light signal generator comprises a total internal reflection prism (TIR prism) adapted to steer an incident light beam with an angle, the angle being preferably comprised between 20° and 40°. [0027] the light signal generator comprises a total internal reflection prism (TIR prism) with a digital micromirror device (DMD) imager. [0028] the first and second optical lines are arranged so that the intersection between the first area and the second area is void. [0029] the first and second optical lines are arranged so that the second area surrounds the first area. [0030] a main direction of light conveyance is defined for the waveguide, and the second optical line and the waveguide are arranged so that the light in the second optical line propagates along a general direction which is perpendicular to the said main direction of light conveyance of the waveguide. [0031] the said waveguide is made from translucent material, preferably from a transparent material. [0032] the first optical line is a folded line. [0033] the second optical line is an unfolded line. [0034] the light signal generator comprises a light source, [0035] the light signal generator comprises a light source and the first optical line is comprising: a collimating lens, the total internal reflection prism (TIR prism), a digital micromirror device (DMD) imager, and a projecting lens. [0036] the light signal generator comprises a light source and

the light source is a fibered source, [0037] the light signal generator comprises a light source and the light source is an electroluminescent diode, more particularly a light-emitting diode (LED) or LED derivatives (e.g. OLED). [0038] the light source of the second light signal is environment of the wearer. [0039] the viewing apparatus is part of or consists in or is associated with googles, [0040] the viewing apparatus is an apparatus for optogenetics uses. [0041] the viewing apparatus is a virtual reality apparatus. [0042] the second optical line is parallel to the gaze direction of the eye of the wearer. [0043] the field of view of the projecting system extends between 15° and 25° along a first direction and between 25° and 35° along a second direction perpendicular to the first direction.

[0044] The specification also relates to a method for projecting light signal in an eye of the wearer of a viewing apparatus comprising a first optical line comprising (i) a projecting system adapted to project a first light signal on a first area of the retina of an eye of the said apparatus wearer, (ii) a light signal generator generating the said first light signal to be projected, and (iii) a waveguide comprising an input coupler and an exit coupler, the waveguide being adapted to convey light from the input coupler to the exit coupler. According to preferred embodiment, the viewing apparatus used in the method of the Invention further comprises a second optical line comprising an optical system adapted to project a second light signal of the environment on a second area of the retina of the eye of the wearer.

[0045] According to particularly preferred embodiment, the method comprises simultaneously the step of projecting a first light signal with the first optical line and of projecting an environment light signal with the second optical line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0046] The invention will be better understood on the basis of the following description which is given in correspondence with the annexed figures and as an illustrative example, without restricting the object of the invention. In the annexed figures:

[0047] FIG. 1 shows schematically an example of viewing apparatus in a context of use by a wearer, and

[0048] FIG. 2 is a detailed view in section of the viewing apparatus of FIG. 1.

DETAILED DESCRIPTION OF SOME EMBODIMENTS

[0049] A viewing apparatus **10**, a wearer **12** and a part of the environment **14** of the wearer **12** are represented on FIG. 1. The part of the environment comprises trees **14A**.

[0050] A viewing apparatus **10** is an apparatus used by the wearer **12** of said apparatus **10** for viewing the part **14** of the environment.

[0051] An example of specific viewing apparatus **10** is further detailed on FIG. 2.

[0052] A first direction parallel to axis X and called “first direction X” is defined. A second direction parallel to axis Y and perpendicular to axis X is defined and called “second direction Y”. A third direction parallel to axis Z and perpendicular to axis X and Y is defined, and called “third direction Z”.

[0053] The viewing apparatus **10** comprises two optical lines, a first optical line **16** and a second optical line **18**.

[0054] An optical line is the set of elements along the optical line that link a source of light signal to the eye **19** of the wearer **12**.

[0055] According to preferred embodiments of the invention, said light signal is an image.

[0056] The first optical line **16** is a folded line.

[0057] It is understood by “folding line” that elements are arranged in such a manner that a light beam along the first optical path has a polygonal chain **16A**.

[0058] The first optical line **16** comprises a light signal generator **20**, a projecting system **22**, and a waveguide **24**.

[0059] The light signal generator **20** is adapted to generate the first light signal to be projected.

[0060] The light signal generator **20** comprises a light source **26**, a collimating lens **28**, a total internal reflection (TIR) prism **30** and a digital micromirror device (DMD) whose reference sign is 32.

[0061] In some cases, the wording DMD imager can be found for element **32**.

[0062] According to the example of FIG. 2, the path of the light in the first optical line **16** follows the path of elements **26**, **28**, **30**, **32**, **30**.

[0063] The light source **26** comprises a light-emitting element **34** adapted to generate light.

[0064] The generated light is selected according to the use of the viewing apparatus of the Invention. For example, in the case of the use of the viewing apparatus by a person equipped with a retina implant with photodiodes, the light-emitting element will be adapted so as to generate a light intensity that stimulates the photodiodes of the implant. Likewise, in the case of the use of the viewing apparatus by a person who has benefitted from optogenetic treatments, the light-emitting element will be adapted to provide a light intensity at wavelengths specific to this application, which stimulate the treated zone.

[0065] The light-emitting element **34** is therefore chosen with in particular a spectral range and/or an intensity corresponding to the application envisaged. The light-emitting element **34** may for example be a laser source or a non-coherent light source.

[0066] According to preferred embodiment, the light-emitting element **34** is a non-coherent light source.

[0067] For example, the light-emitted element **34** comprises an electroluminescent diode, more particularly a light-emitting diode (LED) or LED derivatives (e.g. OLED).

[0068] In the specific example, the light source **26** also comprises a light fiber **36** to convey light to the light emitting element **34**.

[0069] More precisely and as a particular example, the light source **26** is an electroluminescent diode (not shown) which is followed by an optical fiber **36**.

[0070] In other words, the light source **26** is a fibered LED.

[0071] According to preferred embodiment, the light-emitting element **34** is adapted to provide a light at wavelength able to depolarize cell that includes a light-activated ion channel polypeptide. Exemplary wavelengths of light that may be used to depolarize a cell expressing a light-activated ion channel polypeptide, include wavelengths from at least about 365 nm, 385 nm, 405 nm, 425 nm, 445 nm, 465 nm, 485 nm, 505 nm, 525 nm, 545 nm, 565 nm, 585 nm; 590 nm, 605 nm, 625 nm, 645 nm, 665 nm, 685 nm; and 700 nm, including all wavelengths therebetween. In preferred embodiments, light-emitting element **34** is adapted to provide a light intensity at wavelengths in a range of 365 nm to 700 nm, preferably ranging from 530 nm to 640 nm, preferably ranging from 580 nm to 630 nm, preferably ranging from 530 nm to 610 nm, and more particularly the light wavelength is about 595 nm.

[0072] As used herein, the term “about” when used in conjunction with numerical amount, means plus or minus 10% of that numerical amount.

[0073] The collimating lens **28** is adapted to collimate incident light beams **31** coming from the light source **26**.

[0074] The light beams **31** emerging from the collimating lens **28** are substantially parallel to each other along the first direction X.

[0075] For example, the collimating lens **28** has a focal length selected in the range of 2 millimeters (mm) and 20 mm, more specifically in the range of 3 mm and 18 mm, even more particularly the collimating lens **28** has a focal length of about 5.9 mm.

[0076] A total internal reflection prism is often named TIR prism by using the acronym of Total Internal Reflection. TIR prisms are well known in the art.

[0077] The TIR prism **30** is adapted to steer an incident light **31** beam with an angle α , named in what follows the “steering angle α ”.

[0078] The TIR prism **30** comprises an input face **30A**, an output face **30B**, a reflection face **30C**, an incident prism **301** and an output prism **302**.

[0079] The incident face **30A** receives the incident light beam.

[0080] The light beam outputs the TIR prism **30** by the output face **30B**.

[0081] The input face **30A** and the output face **30B** are for example perpendicular to each other.

[0082] The steering angle α is the angle between the light beam emerging from prism **301** and the normal to the digital micromirror device **32**.

[0083] The steering angle α is for example comprised between 20° and 40° .

[0084] By definition, a comprised between the values A and B means that, on the one hand, a is superior or equal to the value A and that a is inferior or equal to the value B.

[0085] In a specific embodiment, the steering angle α is equal to 24° .

[0086] The incident face **30A** belongs to the input prism **301** and the output face **30B** belongs to the output prism **302**.

[0087] The reflection face **30C** belongs to the input prism **301**.

[0088] Input prism **301** and an output prism **302** have complementary faces.

[0089] A tiny gap **40** is defined between the input prism **301** and the output prism **302**. The tiny gap **40** is defined between their complementary faces.

[0090] A digital micromirror device **32** is often named DMD by using the acronym of Digital Micromirror Device.

[0091] DMD are well known in the art. DMD comprises an array (e.g. 608×684 array or 912×1140 array) of mirrors that can switch regularly (e.g. every 0.35 millisecond (ms)) between two discrete angular positions named ON and OFF, with the ON position reflecting the incoming light towards the target. Processed events are encoded by setting the corresponding mirror ON.

[0092] The digital micromirror device **32** is arranged in the optical path of the light beam reflected by the reflected face **30C** of the input prism **301** and adapted to reflect at least a part of the light beam toward the output prism **302**.

[0093] A digital micromirror device **32** enables or disables a certain number of pixels in order to form the light signal to be projected.

[0094] The digital micromirror device **32** comprises a plurality of mirrors that forms arrays of micromirrors.

[0095] Each mirror has at least two positions.

[0096] In a first position, the mirror reflects light beams toward the output prism **302** such that the light is transmitted to the output prism **302** and emerges from the output face **30B**.

[0097] In a second position, the mirror reflects light beams in such a way that no light is outputted from the output face **30B**.

[0098] The projecting system **22** is adapted to project a first light signal on a first area **23** of the retina **19A** of an eye **19** of the wearer **12** of the viewing apparatus **10**.

[0099] In the example, the projecting system **22** is a projecting lens **42**.

[0100] The field of view of the projecting system extends between 15° and 25° along a first direction and between 25° and 35° along a second direction perpendicular to the first direction.

[0101] The waveguide **24** comprises a body **24A**, an input coupler **24B** and an exit coupler **24C**.

[0102] The waveguide **24** is adapted to convey light from the input coupler **24B** to the exit coupler **24C**.

[0103] According to preferred embodiment, all or part of the body **24A** of the waveguide **24** is made from a translucent material.

[0104] In the following description, the term “translucent” refers to a light-transmitting body with a light transmission rate between 5% and 100%.

[0105] More preferably, all or part of the body **24A** is made from a transparent material.

[0106] The term “transparent” refers to light-transmitting body with a light transmission rate greater than 95%, for example close to 100%. In particular, such a “transparent” body transmits light by refraction and through which objects are clearly visible.

[0107] Light in the waveguide **24** is composed of electromagnetic waves whose wavelength is between 500 nanometers (nm) and 700 nm.

[0108] Moreover, the body **24A** of the waveguide is a parallelepiped.

[0109] In the example of FIG. 2, a main direction is defined for the waveguide **24** defined by the length of the body **24A** of the waveguide **24**. Such main direction is a main direction for light conveyance by the waveguide **24**.

[0110] An inside face **24D** and an exterior face **24E** is defined for the body. The inside face **24D** and the exterior face **24E** are normal to the second direction Y.

[0111] The inside face **24D** is intended to be directed toward the projector system **22** of the viewing apparatus and the eye **19** of the wearer when the viewing apparatus is worn.

[0112] The exterior face **24E** is intended to be directed toward the part of the environment **14**.

[0113] In the particular example of FIG. 2, the main direction is along the first direction X.

[0114] The input coupler **24B** and the output coupler **24C** are arranged on the inside face **24D** of the body **24A**.

[0115] For example, the input coupler **24B** is arranged at an extremity of the inside face **24D** and the output coupler **24C** is arranged at the other extremity of the inside face **24D** along the main direction.

[0116] The input coupler **24B** is facing the projecting lens **42**.

[0117] The input coupler **24B** forms an incidence zone of the waveguide **24** whereby the light beams exiting the projector system **22** enters.

[0118] This input coupler **24B** comprises a surface treatment of a part of the inside face **24D** of the body **24A**.

[0119] The input coupler **24B** is made from a thin film in polymer.

[0120] By definition, a thin film is a film whose thickness is inferior to 10 micrometers (μm).

[0121] By definition, an object is made in polymer when the content of the material of the object in polymer is superior or equal to 80%.

[0122] According to special embodiment, said polymer is silver halide.

[0123] The exit coupler **24C** is facing the eye **19** of the wearer **12**.

[0124] The exit coupler **24C** forms an output zone of the waveguide **24**.

[0125] This output coupler **24C** comprises a surface treatment of a part of the inside face **24D** of the body **24A**.

[0126] The exit coupler **24C** is made from a thin film in polymer.

[0127] The second optical line **18** is an unfolded line.

[0128] It is understood by unfolding line that the elements of the second optical line **18** are aligned such as a light beam along the second optical line **18** has a substantially straight path **18A** compared to the path of the light beam along the first optical line **16**.

[0129] In particular, the second optical line **18** is parallel to the gaze direction of the eye **19** of the wearer. For example, the second optical line is parallel to the second direction Y.

[0130] The second optical line **18** comprises an optical system **44** adapted to project/transmit a second light signal of the environment on a second area **46** of the retina **19A** of the eye of the wearer.

[0131] According to special embodiment, the optical system **44** is part of the waveguide **24**, named **25**. It can further be independent from said waveguide **24**. Alternatively, the second optical line **18** arrives directly from the environment to the eye of the wearer.

[0132] The limits of this part **25** of the waveguide **24** are defined by the optical field of view angle of the wearer **12**.

[0133] More precisely, the part **25** is the intersection of the waveguide with the field of view angle

of the wearer **12**.

[0134] The part **25** comprises the output coupler **24C**.

[0135] The second optical line **18** and the waveguide **24** are arranged so that the light in the second optical line **18** propagates along a general direction which is perpendicular to the main direction of the waveguide **24**. For example, the general direction is along the second direction Y.

[0136] The two optical lines **16**, **18** are arranged so that the intersection between the first area **23** and the second area **46** of the retina **19A** is void. The space between the first and second areas **23**, **46** is called “void area **48**”.

[0137] Alternatively or in combination with the previous arrangements, the two optical lines **16**, **18** are arranged so that the second area **46** surrounds the first area **23**.

[0138] According to special embodiment, the viewing apparatus **10** comprises at least one lens to correct vision defects, such as for example eye glass lens. Such lens to correct vision defects are preferably located on optical line **16** between output coupler **24C** and the eye of the wearer and/or optical line **18** before eye of the wearer.

[0139] According to special embodiment, the viewing apparatus **10** comprises at least one optical filter, for example colored optical filter. Such a filter can be located along the optical line **16** (for example between the collimating lens **28** and the total internal reflection prism **30**, and/or between the total internal reflection prism **30** and the digital micromirror device **32**, and/or between the DMD and the projecting lens **42**, and/or between the projecting lens and the waveguide **24**) and/or along the optical line **18**.

[0140] The operating of the viewing apparatus **10** is now described in reference to a method for projecting light signals on an eye of the wearer.

[0141] According to special embodiment, the viewing apparatus **10** is in the form of a pair of spectacles.

[0142] The method comprises a step of providing, a step of positioning and two steps of projecting.

[0143] During the step of providing, the viewing apparatus **10** is provided to the wearer **12**.

[0144] As a specific example, an eye professional gives the viewing apparatus **10** to the wearer **12**.

[0145] At the step of positioning, the wearer takes the viewing apparatus and positions it on his head in front of his eyes.

[0146] For instance, when the viewing apparatus **10** comprises a fixing band, the wearer **12** uses the fixing band to have the viewing apparatus **10** fixed on his head.

[0147] The two steps of projecting are achieved simultaneously.

[0148] According to the first step of projecting, a first light signal is projected with the first optical line **16** on the first area **23** of the retina **19A**.

[0149] The first light signal to be projected is generated by the light signal generator **20**.

[0150] More precisely, incident light beams **31** generated from the fibered LED **26** are transmitted to the TIR prism **30**.

[0151] Light beams **31** are reflected by the reflection face **30C** of the input prism **30** toward the micromirror device **32**.

[0152] In function of the position of the micromirrors of the micromirror device **32**, the light beam is split into a selected part, reproducing the light signal to be projected, and a discarded part.

[0153] The selected part is projected through the output prism **302**.

[0154] The light beam emerging from the output face **30B** is transmitted to the projecting lens **42**.

[0155] The light beam exiting the projected lens **42** is transmitted to the waveguide **24** through the input coupler **24B**.

[0156] The body **24A** of the waveguide **24** conveys the light beam to the output coupler **24C** whereby the light beam is exiting the waveguide **24**.

[0157] The light beam exiting the output coupler is projected in the first area **23** of the retina of the wearer **12**.

[0158] Simultaneously, the method comprises the second step of projecting.

[0159] According to the second step of projecting, an environment light signal (i.e. second light signal) is projected with the second optical line **18**.

[0160] Light beams from the environment passes through the body **24A** of the waveguide **24** and reach the second area **46** of the retina **19A**.

[0161] The light beams from the environment are transmitted into the body **24A** by the exterior face **24E**, cross the body **24A** and come out the waveguide by the output coupler **24C** and are projected onto the second area **46** of the retina **19A**.

[0162] The waveguide **24** will keep the peripheral vision of the patient intact while enabling the light signal generator **20** to be placed away from the wearer's visual field.

[0163] Using a fibered LED **26** together with a reflection prism **30** also makes it possible to have the LED away from the projecting system thus reducing heating problems for the wearer **12**.

[0164] Moreover the second light signal projected with the second optical line **18** completes light signal projected with the first optical line for rebuilding a part of the environment.

[0165] Moreover, thanks to the second optical line **18**, it is ensured that the second light signal is projected on a second area **46** of the retina **19A** which does not have any overlap area with the first area **23** of the retina **19A**.

[0166] Moreover, the viewing apparatus **10** of the invention enables to obtain a device independent of pupil size especially if pupil size is above 3 millimeters (mm) in diameter.

[0167] The viewing apparatus **10** also allows delivering the necessary irradiance to the retina **19A** of the wearer **12**.

[0168] The viewing apparatus **10** is also compact. In particular, as the first optical line **16** is a folded line, it is possible to bring the elements of the first optical line **16** closer to the waveguide **24**.

[0169] Thus the dimension of the viewing apparatus along the second direction is reduced compared to known viewing apparatus.

[0170] Such viewing apparatus also provides the wearer with a field of view of $20^{\circ} \times 30^{\circ}$ thanks to the use of the whole apparatus **10** and notably the TIR prism **30** with the digital micromirror device **32**

[0171] The viewing apparatus **10** does not obscure the patient's visual field.

[0172] According to a specific embodiment, the projecting system (**22**) and/or the light signal generator (**20**) are mounted on the frame of a pair of spectacles.

[0173] The viewing apparatus **10** may be used in the field of vision restoration using vision prostheses such as retinal implants.

[0174] According to a specific embodiment, the viewing apparatus **10** is an apparatus adapted for optogenetics.

[0175] The apparatus **10** defined above may be used for the subjects (i.e. wearer) suffering of photoreceptor loss or degeneration, such as in case of retinitis pigmentosa (RP) or macular degeneration (MD). As mentioned above, these affections diminish visual acuity, diminishes light sensitivity, or result in blindness of a part of the field of view of the subject.

[0176] As explained above, some therapies consist in stimulating transfected cells of the retina and/or retinal implants with a light beam.

[0177] The first area **23** of the retina on which the light signal with the first optical line **16** is intended to be projected comprise cells, more specifically retinal cells, that have been genetically modified for expressing light-gated ion channel protein for optogenetic therapy.

[0178] According to special embodiment, said light-gated ion channel protein is chosen in the group consisting of Chrimson, ChrimsonR, (WO2013/71231), ChrimsonR-tdT (WO2017/187272), Catch, Channelrhodopsin (US20140121265, U.S. Pat. No. 8,906,360), and melanopsin and derivatives thereof.

[0179] The first area **23** of the retina on which the light signal with the first optical line **16** is intended to be projected comprises retinal implant having to be stimulated.

[0180] The area of the retina is stimulated by a light beam reproducing light signal, more specifically an image, of the environment.

[0181] The projected light signal rebuilds the lost field of view due to the photoreceptor loss or degeneration in function of the gaze direction of the subject.

[0182] The light beam obtained with the first optical line and stimulating the parts of the retina is obtained by the method explained above.

[0183] In the detailed embodiment, the environment light signal, more specifically the image, projected with the second optical line **18** corresponds to the part of the environment that complete the light signal, more specifically the image, projected with the first line in such a way that the wearer sees a rebuilt light signal, more specifically the image, of the environment.

[0184] Moreover, the first area **23** of the retina is stimulated with the required light characteristics independent of the pupil size.

[0185] In such case, it is to be noted that thresholds of light intensity (maximum and minimum) are given by phototoxicity standards and are further analyzed in literature relevant to ophthalmology or to the application of light stimulation for an optogenetic therapy (Yan et al. 2016; Delori, Webb, and Sliney 2007; Sliney et al. 2005). For example, for a light with a wavelength of 595 nm, [0186] the maximum light intensity at the retina is 7 mW/mm² (ISO 15004-2 2007; ISO 62471 2006), and [0187] At the cornea (anterior segment), the maximum light intensity is 32 mW over any 1 mm diameter disc (ISO 15004-2 2007).

[0188] In addition, the retinal radiant exposure limit when taking into account the luminance dose restriction is 6.6 J.Math.cm^{sup.-2} over 48 hours (luminance dose restriction, ANSI Z136.1 2014).

[0189] According to another particular example, the viewing apparatus is adapted for a virtual reality apparatus. Head-Mounted displays are used for example used for augmented reality, for virtual reality or for movie display.

[0190] In such case, the light signal projected with the second optical line **18** generally has no specific link with the environment.

[0191] The viewing apparatus and the method disclosed in relation to the viewing apparatus **10** can be used for virtual reality or optogenetic applications.

Claims

1. A viewing apparatus comprising: a projecting system adapted to project a first light signal on a first area of the retina of an eye of a wearer of the viewing apparatus, a light signal generator generating the first light signal to be projected, and a waveguide comprising an input coupler and an exit coupler, the waveguide being adapted to convey light from the input coupler to the exit coupler.
2. The viewing apparatus according to claim 1, wherein the waveguide is made in polymer.
3. The viewing apparatus according to claim 1, wherein the waveguide is a polymer film.
4. The viewing apparatus according to claim 1, wherein the waveguide has a thickness less than 10 micrometers.
5. The viewing apparatus according to claim 1, wherein the projecting system, the light signal general and the waveguide belongs to a first optical line of the viewing apparatus, the viewing apparatus further comprises a second optical line comprising an optical system adapted to project a light signal of the environment on a second area of the retina of the eye of the wearer.
6. The viewing apparatus according to claim 1, wherein the waveguide is translucent.
7. The viewing apparatus according to claim 1, wherein the light signal generator comprises a total internal reflection prism adapted to steer an incident light beam with an angle.
8. The viewing apparatus according to claim 7, wherein the angle is comprised between 20° and 40°.
9. The viewing apparatus according to claim 5, wherein the two optical lines are arranged so that

the intersection between the first area and the second area is void.

10. The viewing apparatus according to claim 5, wherein the two optical lines are arranged so that the second area surrounds the first area.

11. The viewing apparatus according to claim 5, wherein a main direction of light conveyance is defined for the waveguide and wherein the second optical line and the waveguide are arranged so that the light in the second optical line propagates along a general direction which is perpendicular to the main direction of light conveyance of the waveguide.

12. The viewing apparatus according to claim 5, wherein the first optical line is a folded line.

13. The viewing apparatus according to claim 5, wherein the second optical line is an unfolded line.

14. The viewing apparatus according to claim 13, wherein the light signal generator comprises a light source and the first optical line a collimating lens, the total internal reflection prism, a digital micromirror device and a projecting lens.

15. The viewing apparatus according to claim 1, wherein the light signal generator comprises a light source, the light source being at least one of a fibered source and an electroluminescent diode.

16. The viewing apparatus according to claim 1, wherein the viewing apparatus is an apparatus for optogenetics uses.

17. The viewing apparatus according to claim 5, wherein the second optical line is parallel to the gaze direction of the eye of the wearer.

18. The viewing apparatus according to claim 1, wherein the field of view of the projecting system extends between 15° and 25° along a first direction and between 25° and 35° along a second direction perpendicular to the first direction.

19. A method for projecting a light signal in an eye of the wearer of a viewing apparatus comprising: a projecting system adapted to project a first light signal on a first area of the retina of an eye of a wearer of the viewing apparatus, a light signal generator generating the first light signal to be projected, and a waveguide comprising an input coupler and an exit coupler, the waveguide being adapted to convey light from the input coupler to the exit coupler.

20. The method according to claim 19, wherein the viewing apparatus further comprises a second optical line comprising an optical system adapted to project a light signal of the environment on a second area of the retina of the eye of the wearer, the method comprising simultaneously projecting a light signal with the first optical line and projecting an environment light signal with the second optical line.
